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Roll number: **CB.SC.U4CSE23022**

Gopher-AI
Model Test & Evaluation

Framework-aligned evaluation report | 15 October 2026

Evaluated checkpoint: **gopher-ai_v1.pt**
Task: detect five road-damage categories in dashcam images.

This report contains fresh measurements from the supplied model and local RDD2022 data. It adapts the April 2024 Test and Evaluation of Artificial Intelligence Models framework to an object detector. The framework provides strategy-level guidance, not a fixed test recipe or universal pass score [1, printed pp. 3, 19, 41].

Primary result	Measured value
Test images / annotated objects	5,076 / 8,325
mAP@0.50	49.04%
mAP@0.50:0.95	24.50%
Detection precision / recall at confidence 0.25	56.41% / 47.90%
Detection F1 at confidence 0.25	51.81%
Image-level damage-presence accuracy	84.75%
Warm single-image latency, mean / p95	16.20 / 25.36 ms

Evaluation outcome

At the fixed operating threshold, the model correctly detected 3,988 annotated objects and missed 4,337. These measurements support further model development and supervised review; they do not establish autonomous road-safety readiness. No stakeholder acceptance thresholds or live deployment measurements were supplied.

All percentages are measured on the stated sample and configuration. Confidence intervals, subgroup differences, stress-test degradation, and untested areas are recorded in the following pages.

02 / Test strategy and framework mapping

The intended use is to assist a person reviewing road-surface images. Incorrect alarms create review work; missed damage can leave maintenance needs undetected. Model-level testing is performed here; a complete vehicle, application, camera, and operator workflow is outside this experiment [1, pp. 4, 56-57].

Framework area (printed page)	Implementation in this evaluation
Correctness (9, 45)	Full labeled test split; AP, fixed-threshold detection precision/recall/F1; secondary binary image accuracy and ROC-AUC.
Bias / representativeness (11, 16)	Class counts, country-prefix composition, country-specific detection results, and train/test distribution differences.
Robustness / test design (12, 20, 23)	Seeded 300-image sample; lighting, blur, noise, JPEG degradation; a complete 2 x 2 darkness-by-blur comparison.
Metamorphic testing (22)	Horizontal flip with transformed ground-truth boxes and prediction consistency after reversing the flip.
Uncertainty (14)	2,000 image-level bootstrap resamples for metric confidence intervals; confidence reliability and Brier score.
Latency / resilience (17, 13)	100 warm batch-one timings, tail latency, memory; 20 clean-corrupt-clean recovery probes.
Interpretability (10)	Local 6 x 6 occlusion sensitivity study of one correctly detected pothole.
Data / documentation (31-37, 59-63)	Image/label audit; exact-duplicate split checks; checksums, manifests, model/data facts, scripts and machine-readable results.

Controls and assumptions

Weights were frozen throughout testing. No retraining or test-set threshold selection was performed. Confidence 0.25 and matching IoU 0.50 were fixed in advance for the operating-point analysis. AP uses scores down to 0.001 and IoUs 0.50 through 0.95. NMS IoU is 0.70 and maximum detections per image is 300.

The primary split is the existing test folder. Filename and exact-content checks establish no observed exact overlap, but do not prove historical isolation: earlier evaluation outputs exist, and prior use for model selection cannot be ruled out. Near-duplicate scenes and route-level overlap were not assessed.

03 / Data quality and representativeness

Split	Images	Objects*	Backgrounds	Background share
Train	16,726	40,197	1,000	5.98%
Val	5,113	8,422	1,837	35.93%
Test	5,076	8,325	1,790	35.26%

*Objects counted by this audit exclude a malformed zero-width training annotation. Duplicate annotation rows are counted and flagged; source labels were not edited.

Class	Training objects	Validation objects	Test objects
Longitudinal crack	15,358	3,261	3,293
Transverse crack	6,738	1,422	1,311
Alligator crack	6,855	1,430	1,397
Other corruption	6,840	1,397	1,419
Pothole	4,406	912	905

Audit findings

All 5,076 test images decoded successfully and had corresponding label files; no invalid label rows, duplicate annotation rows, orphan labels, or exact duplicate test images were found. Pairwise split comparisons found zero shared filenames and zero shared SHA-256 image hashes.

The training split has one zero-width annotation (Japan_001265, row 4), two images with duplicate annotation rows (Japan_006916 and Japan_011427), and one excess exact duplicate image. The validation split has one image with duplicate annotation rows (Japan_006536). These are data-quality findings, not automatic evidence of model failure.

Representation limits

The local test set contains country prefixes for Czech Republic, India, Japan, Norway, and United States. Country is inferred from filenames, not independently verified capture metadata. The repository documents filtering drone/motorbike images, retaining 1,000 training background images, and resizing training/validation images. This audit verifies the current files, not the historical execution of every preprocessing step.

Training backgrounds comprise 5.98%, versus 35.26% in testing. This large sampling difference matters for false alarms. The data audit checks syntax and exact content; expert annotation correctness, privacy review, scene diversity, weather, night coverage, and deployment representativeness remain unverified.

04 / Correctness on the full test split

Detection metrics are appropriate because the model must both localize an object and assign its damage category. For fixed-threshold results, predictions are processed from highest to lowest confidence and matched one-to-one to an unused same-class ground-truth box at IoU >= 0.50.

Class	GT	TP / FP / FN	Precision	Recall	F1
Longitudinal crack	3,293	1356 / 1050 / 1937	56.36%	41.18%	47.59%
Transverse crack	1,311	525 / 509 / 786	50.77%	40.05%	44.78%
Alligator crack	1,397	804 / 544 / 593	59.64%	57.55%	58.58%
Other corruption	1,419	1005 / 724 / 414	58.13%	70.82%	63.85%
Pothole	905	298 / 255 / 607	53.89%	32.93%	40.88%
All (micro)	8,325	3988 / 3082 / 4337	56.41%	47.90%	51.81%

Threshold-swept average precision

Class	AP@0.50	AP@0.50:0.95
longitudinal crack	43.15%	22.05%
transverse crack	39.59%	16.57%
alligator crack	59.50%	29.86%
other corruption	67.16%	39.34%
Pothole	35.82%	14.68%
Mean over five classes	49.04%	24.50%

Precision = TP / (TP + FP); recall = TP / (TP + FN); F1 = 2PR / (P + R). AP integrates the precision-recall curve; mAP averages class AP. AP@0.50:0.95 also averages ten localization thresholds. The Ultralytics AP calculation uses its built-in matching method and multi-label validation NMS; the fixed-threshold table uses the explicitly defined confidence-ordered matching rule.

Secondary image-level classification

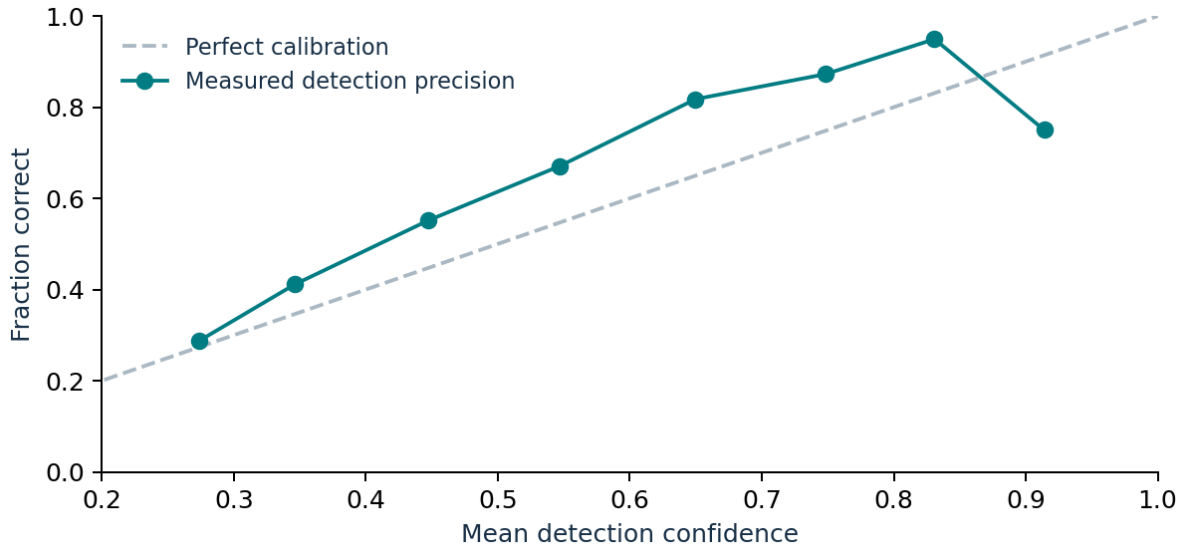
Damage presence is positive if an image has any annotation; prediction is positive if any detection has confidence >= 0.25. TP=2,846, FP=334, TN=1,456, FN=440. Accuracy is **84.75%**, ROC-AUC is **0.9137**, and specificity is 81.34%. This task ignores location and damage type; it must not be presented as object-detection accuracy.

05 / Uncertainty and confidence reliability

Confidence intervals describe sampling variability for the fixed model. Entire images, including their ground truths and detections, were resampled with replacement 2,000 times (seed 23022). The table reports the 2.5th and 97.5th percentiles. The model was not retrained during resampling.

Metric	Point estimate	95% image-bootstrap interval
Detection precision	56.41%	55.31% - 57.48%
Detection recall	47.90%	46.40% - 49.44%
Detection F1	51.81%	50.68% - 52.94%
Image-level accuracy	84.75%	83.75% - 85.72%

These intervals assume test images are exchangeable. Correlated frames or shared routes can make the intervals too narrow. They do not capture new-country uncertainty, labeling errors, model-training variability, or all aleatoric/epistemic uncertainty. This is evaluation bootstrapping, distinct from the training-resampling example in framework p. 23.



For 7,070 retained detections, expected calibration error is **0.0908** and detection Brier score is **0.2111**. Ten equal-width bins span [0,1]; only occupied bins contribute. A detection is correct only if it matches a same-class object at IoU >= 0.50 under the stated matching rule.

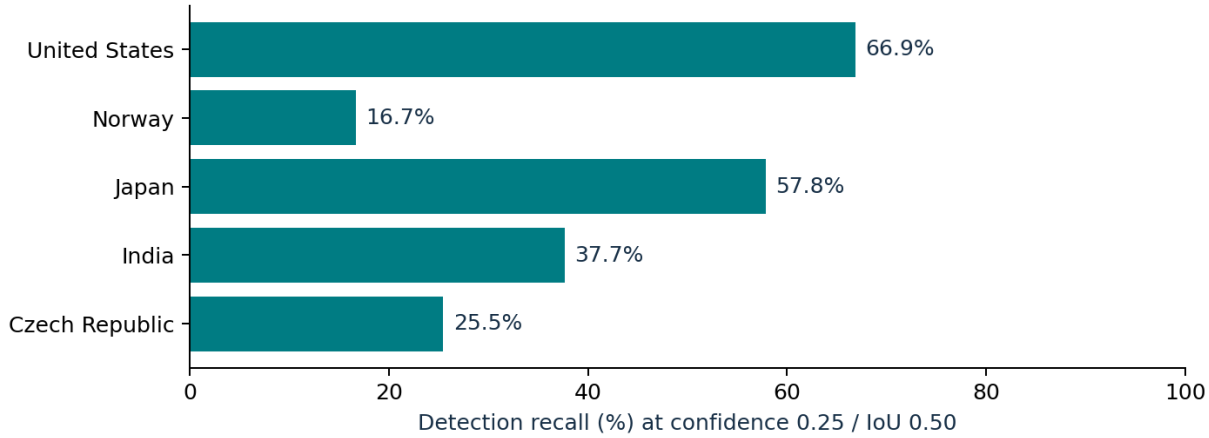
The 0.90-1.00 confidence bin contains only four detections, so its final downward point is highly uncertain. Bin sample counts are retained with the results; the chart should not be read as a smooth population calibration curve.

The binary image-level Brier score is 0.1907, using maximum detection confidence as a damage-presence score. That score is a proxy, not a separately trained image probability. No calibration was fitted on the test set. Low calibration error for retained detections does not account for missed objects or establish overall uncertainty quality.

06 / Subgroup performance and operating trade-offs

Country-prefix groups provide a limited view of representation bias. The table uses the same confidence and matching thresholds as the primary fixed-threshold analysis. Differences may reflect class mix, image conditions, annotation practices, or geographic domain effects; they do not establish a causal explanation or demographic fairness.

Country prefix	Images	Objects	Precision	Recall	F1
Czech Republic	436	267	46.90%	25.47%	33.01%
India	1,166	1,168	44.13%	37.67%	40.65%
Japan	1,524	3,617	60.62%	57.84%	59.20%
Norway	1,197	1,595	50.28%	16.68%	25.05%
United States	753	1,678	57.60%	66.87%	61.89%



The highest-to-lowest country recall gap is **50.19 percentage points**. All groups contain multiple images, but coverage of each class within each environment is not guaranteed. No protected-person attributes are available.

Confidence-threshold sensitivity

Confidence	Precision	Recall	F1	FP / FN
0.10	32.16%	63.26%	42.64%	11,106 / 3,059
0.25	56.41%	47.90%	51.81%	3,082 / 4,337
0.50	79.67%	26.40%	39.66%	561 / 6,127
0.75	91.75%	6.94%	12.91%	52 / 7,747

This sweep illustrates trade-offs; it was not used to select a best test threshold. Any deployment threshold should be chosen on separate validation data using agreed costs of misses and false alarms.

07 / Robustness and metamorphic testing

A uniform random sample of 300 test images (seed 23022, without replacement) was reused for every condition. Images were resized to a 512-pixel longest side before corruption, then square-letterboxed for inference. Each condition uses the same ground truth, except horizontal-flip boxes are mirrored. Results below are micro detection metrics at confidence 0.25 / IoU 0.50.

Condition	Precision	Recall	F1	F1 change (pp)
Clean reference	50.90%	40.96%	45.39%	+0.00
Brightness x0.75	51.04%	40.96%	45.44%	+0.05
Brightness x0.50	54.65%	40.33%	46.41%	+1.02
Brightness x1.50 (clipped)	48.31%	35.55%	40.96%	-4.43
Gaussian blur: 5x5, sigma 1.5	61.60%	16.01%	25.41%	-19.98
Gaussian noise: sigma 15/255	54.37%	23.28%	32.61%	-12.79
JPEG quality 30	51.80%	23.91%	32.72%	-12.67
Brightness x0.50 + blur	56.15%	15.18%	23.90%	-21.50
Horizontal flip	52.00%	40.54%	45.56%	+0.17

The sample contains 481 objects. Realized country counts: Czech 19, India 60, Japan 94, Norway 83, United 44. Its clean result is a separate sample/configuration baseline, not directly interchangeable with the full-set AP run. Predict-mode uses single-label NMS.

Pairwise interaction test

Clean, darkness-only, blur-only, and darkness-plus-blur cover all four combinations of two binary factors. This is a complete 2 x 2 factorial test for these factors; it does not cover every pair among all possible environmental variables. The simulated corruptions probe input sensitivity and are not measurements of real night, rain, or fog performance.

Metamorphic relation: horizontal reflection

After flipping predicted boxes back, 312 same-class boxes match the clean predictions at IoU >= 0.50. Clean-detection retention is **80.62%**, symmetric box-consistency F1 is **81.89%**, and 199/300 images have complete box consistency (including images empty in both runs).

The proposed relation is approximate invariance to reflection. It is a diagnostic assumption, not a guarantee that every road scene is semantically identical when mirrored. No adaptive adversarial attack, poisoning experiment, or red-team exercise was performed.

08 / Latency and recovery from perturbation

Measurements used NVIDIA GeForce GTX 1650 (4 GiB), FP32, 512 x 512 input, and four PyTorch CPU threads. Batch-one timings used 10 warm-up images followed by 100 images with CUDA synchronization around each complete prediction call. Confidence was 0.25. The timer includes API overhead, preprocessing, inference, and postprocessing; file decode, initial resize, camera capture, network transport, and UI rendering are excluded.

Latency / resource measure	Measured value
Mean / median prediction-call latency	16.20 / 14.84 ms
95th / 99th percentile latency	25.36 / 35.35 ms
Minimum / maximum latency	10.50 / 39.24 ms
Serial throughput (1000 / mean milliseconds)	61.75 images/s
Mean internal preprocess time	1.55 ms
Mean internal inference time	12.29 ms
Mean internal postprocess time	1.85 ms
Peak CUDA tensor allocation during timed sample	57.03 MiB

The full test validator separately reported 6.92 ms/image of inference with batch size 8. This is an amortized component time, not single-image end-to-end latency. GPU tensor allocation excludes reserved allocator memory and other processes; it is not total device memory use.

Clean-corrupt-clean resilience probe

Twenty input sequences applied the original image, a darkened version (brightness x0.50), then the original image again. **20/20** restored predictions matched their initial outputs (class list exact, confidence absolute tolerance 1e-6, box tolerance 1e-4; NumPy default relative tolerance 1e-5). The mean restoration-check time, including the prediction call, output conversion and comparison, was **18.22 ms**.

This is a stateless feed-forward detector, so the probe tests lingering input effects in the inference process. It does not test crash recovery, sensor reconnection, tracking state, service failover, power interruption, or recovery of a complete deployed system. No operational latency requirement was supplied, so no real-time pass/fail claim is made.

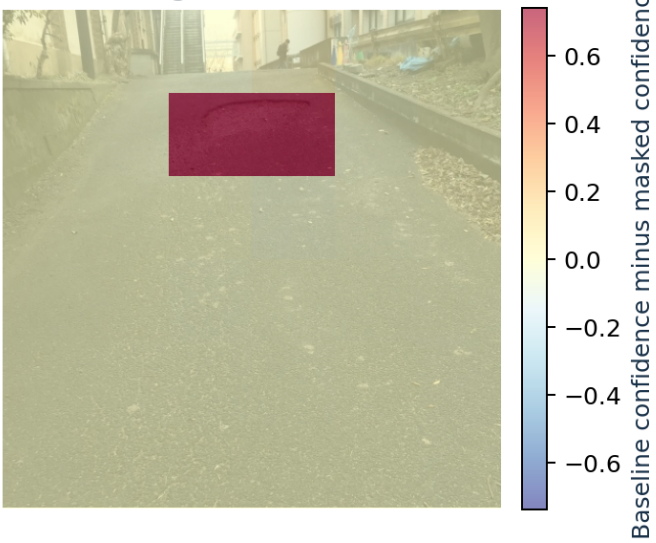
09 / Local interpretability probe

A post-hoc, model-agnostic occlusion study was applied to one selected correct pothole detection. The image was partitioned into a 6 x 6 grid. One cell at a time was replaced with gray value 114, and the model was rerun. The tracked score is the highest pothole confidence at IoU >= 0.50 with the original predicted box; it becomes zero when no such detection remains.

Selected correct pothole detection



Confidence change when each cell is masked



Probe fact	Measured result
Image	Japan_003154.jpg
Selection rule	highest-confidence correctly detected pothole in the random 300-image subset
Baseline confidence	0.7391
Number of occlusion runs	36
Maximum confidence decrease	0.7391
Mean signed confidence decrease	0.0393
Minimum signed confidence decrease	-0.0174

Positive values identify cells whose removal reduced the tracked confidence; negative values identify cells whose removal increased it. A large drop can reflect changed localization as well as changed class confidence. Gray occlusion itself can create an out-of-distribution artifact.

This probe shows local sensitivity, not a causal explanation of learned reasoning. It is neither a global explanation nor proof that the model exclusively uses road-damage features. The selection rule favors a strong correct detection and is not a random estimate of explanation quality. No human explanation-usability study was performed.

10 / Interpretation, limitations and next actions

The model's full-test mAP@0.50 is 49.04%, while mAP@0.50:0.95 is 24.50%. The stricter localization metric is lower, so detecting an approximate region does not imply precise bounding boxes. At confidence 0.25, 52.10% of annotated objects are missed and 18.66% of background-only images receive at least one alarm.

The lowest measured subgroup recall is 16.68% for the Norway prefix. The lowest stress-sample F1 is 23.90% under Brightness x0.50 + blur. These are priorities for investigation, not proof that one geographic or environmental factor causes the errors.

Framework method / concern	Status and reason
Holdout validation	Executed on the existing test split; historical test isolation remains unproven.
Cross-validation / training bootstrap	Not executed. Valid evaluation across training folds requires retraining independent models; subdividing this model's test set would not provide that evidence.
A/B or back-to-back model comparison	Not executed. No agreed task-compatible comparator was selected; the COCO-pretrained model has different labels.
Adversarial testing / red teaming	Not executed. Natural corruption probes do not establish resistance to optimized attacks or data poisoning.
Temporal drift	Not measured. No timestamped production series or newer labeled deployment data was supplied. Train/test composition differences are not temporal drift.
Fairness, field testing and compliance	Not established. Requires relevant stakeholder definitions, metadata, deployment evidence, and review.
Continuous monitoring / human evaluation	Not established. This is a discrete automated model-level test event, with a local sensitivity visualization.

Recommended follow-up

1. Review missed potholes and weak country/class combinations; repair verified annotation defects in versioned data.
2. Choose operating thresholds on separate validation data using agreed miss/alarm costs; preserve an untouched future test set.
3. Collect route-disjoint, labeled deployment images covering intended cameras, roads, weather and light; include verified negative scenes.
4. Re-evaluate new versions, add adversarial and system recovery tests, and monitor field latency, class mix and labeled performance over time.

11 / Model card, reproducibility and references

Model / data fact	Recorded value
Owner / report identity	Eswara Raj M | CB.SC.U4CSE23022
Architecture / learning paradigm	YOLO11n; supervised convolutional object detection; five classes
Input / output	Color images; bounding boxes, damage classes and confidence scores
Unfused parameters / checkpoint size	2,590,815 / 21,254,689 bytes
Checkpoint metadata	Epoch 78 (zero-based; epoch 79); date 2026-09-18T10:50:16.170715
Recorded training configuration	AdamW; initial LR 0.0015; image size 512; batch 16; patience 10; maximum 100 epochs; seed 0; AMP disabled
Evaluation software	Python 3.12.10; PyTorch 2.11.0+cu128; Ultralytics 8.4.90
Source and intended use	Local RDD2022 derivative; academic road-damage analysis with human review
Rights and provenance limit	Repository describes academic/research use; dataset license, consent and full chain of custody were not independently verified.

Version identifiers

Checkpoint SHA-256:
b8295ce24303453a68497c2d0c4539c3d1eabd865192bd6fc345211f37f7b01a

Ordered image-and-label manifest SHA-256:
7cf398cb8a9378af85468e0c1dad2c54fd23c1cc37a06e5e607a4199756e41e1

Reproduce the measurements

Run the included scripts from the repository: audit_data.py, run_baseline.py, analyze_results.py, then run_extended.py. They use the existing weights and local dataset. The evidence bundle includes the scripts, settings, scalar metrics, per-image results, audit findings and hash manifest. Raw predictions and complete dataset records are retained in output/evaluation in the workspace. Report generation uses the bundled PDF environment.

References and page convention

[1] CDAO Assessment and Assurance. Test and Evaluation of Artificial Intelligence Models: What to Consider in a Test & Evaluation Strategy. April 2024. User-supplied PDF, 66 PDF pages. All cited page numbers in this report refer to printed slide numbers; PDF page = printed page + 1. This report paraphrases and adapts the framework.

[2] Local project evidence: README.md, new_train.py, runs/detect/train-10/args.yaml, gopher-ai_v1.pt checkpoint metadata, and current RDD2022 files. The supplied checkpoint matches the local best.pt SHA-256. Training configuration is recorded metadata, not independently reproduced training.

[3] Installed Ultralytics 8.4.90 DetectionValidator and detection metrics implementation, inspected locally; fresh execution results and evaluation scripts accompanying this report.